

The synaptic Package

User Guide

synaptic project

CTAN Community

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Abstract: **synaptic** is a modular $\text{\LaTeX}$ design system for academic articles, books, lecture notes, and technical reports. It requires LuaLaTeX with a KOMA-Script document class. This guide covers installation, options, and usage.

Keywords: synaptic, user guide, LaTeX, LuaLaTeX

arXiv: [2605.54321](https://arxiv.org/abs/2605.54321)

To the elegance of structured typesetting.

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1 Installation

Clone the repository and add `tex/latex/synaptic/` to your TEXINPUTS:

```
git clone https://github.com/lzxc/synaptic
export TEXINPUTS=/path/to/synaptic/tex/latex/synaptic/:$TEXINPUTS
```

Or install via l3build:

```
cd synaptic
lualatex synaptic.ins    # extract .sty files
l3build install          # install to TEXMFHOME
```

2 Options

All options are passed via `\usepackage[key=val,...]{synaptic}`. Runtime changes use `\SynapticSetup{key=val}`.

2.1 Publication Modes (`mode=`)

Value	Description
journal	Academic paper. Submission header, abstract, keywords, arXiv.
book	Monograph or thesis. Chapter headings, running headers, front/back matter.
lecture	Lecture notes. Wide margins, large headings, tcolorbox environments.
notes	Personal notes. Compact layout, minimal extras.

2.2 Colour Themes (`theme=`)

Theme	Primary	Secondary
ocean	Blue (0,106,176)	Dark blue-grey (46,64,83)
graphite	Grey (75,86,99)	Dark grey (45,52,62)
forest	Green (30,115,80)	Dark green (56,68,60)
midnight	Navy (25,55,109)	Deep dark (30,35,50)
paper	Dark grey (55,58,62)	Near-black (45,48,52)

Custom themes: `\SynapticDefineTheme{mytheme}{RGB:...}{RGB:...}`.

2.3 Font Sets (fontset=)

Value	Roman	Sans	Math
auto	Auto-detect	Auto-detect	Auto-detect
xcharter	XCharter	Cabin	XCharter Math
libertinus	Libertinus Serif	Libertinus Sans	Libertinus Math
stix2	STIX Two Text	STIX Two Text	STIX Two Math
lm	Latin Modern	Latin Modern Sans	Latin Modern Math

2.4 Other Options

Key	Type	Description
lang	choice (en, zh)	Language for labels and CJK fonts.
toc	bool (true)	Show table of contents.
toc-layout	choice (top, inline, page)	TOC position.
extra-math	bool (false)	Enable $\mathbb{A}$, $\mathcal{A}$, $\mathfrak{A}$.
fine-breaking	bool (true)	Micro-typographic line breaking.

3 User API

3.1 Metadata Commands

`\SynapticTitle{text}` — document title (replaces `\title`).

`\SynapticAuthor[mark]{name}` — author with optional affiliation mark.

`\SynapticAffiliation[mark]{text}` — affiliation.

`\SynapticEmail{addr}`, `\SynapticKeywords{...}`, `\SynapticArxiv{ID}` — email, keywords, arXiv link.

`\SynapticHeader{text}`, `\SynapticSubHeader{text}`, `\SynapticPreprint{ID}`, `\SynapticDedicated{...}` — custom header fields.

`\SynapticMakeTitle` — render the title page.

3.2 Customisation

`\SynapticSetup{key=val}` — change options after loading.

`\SynapticNewTheorem{env}{Name}` — declare theorem-like environment.

`\SynapticDefineTheme{A}{B}` — custom colour theme.

4 Mathematics

Theorem 4.1 (Euler's Identity). *For any real number x , $e^{i\pi} + 1 = 0$.*

Proof. Follows from the definition of complex exponentials. ■

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

With extra-math: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R} \subset \mathbb{C}$ and $\mathcal{A} \xrightarrow{\sim} \mathfrak{B}$.

5 List and Caption Styling

1. First item with some content to show spacing.
2. Second item
 - Sub-item one
 - Sub-item two
3. Third item concludes the list.

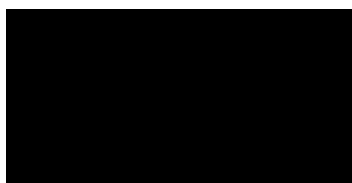


Figure 1. A sample figure with styled caption.

Acknowledgments

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